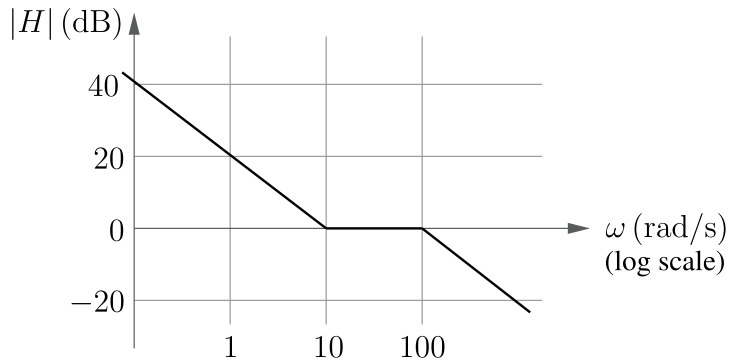
Basic Electronics

Instructor: M. B. Patil, IIT Bombay
Assignment for Week 8

1. Which of the following transfer functions corresponds to the magnitude Bode plot shown in the figure?



(A) $\dfrac{10\,(s+100)}{s\,(s+10)}$ (B) $\dfrac{100\,(s+10)}{s\,(s+100)}$ (C) $\dfrac{10\,s}{(s+10)(s+100)}$ (D) $\dfrac{100\,(1+0.1s)}{(1+0.01s)}$

2. Consider the transfer function, $\dfrac{50\,s}{(s+10)(s+100)}$. What is the phase obtained using the Bode approximation at $\omega = 10\,\text{rad/s}$?

(A) 55° (B) 40° (C) 45° (D) 35°

3. For the transfer function of Q-2, what is the exact phase (i.e., without the Bode approximation) at $\omega = 10\,\text{rad/s}$?

(A) 39° (B) 44° (C) 34° (D) 52°

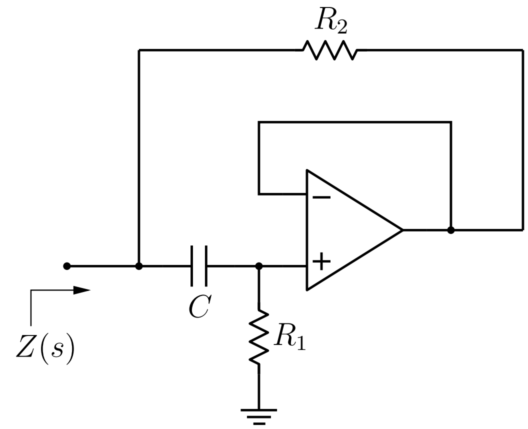
4. For the transfer function of Q-2, what is the phase obtained using the Bode approximation at $\omega = 100\,\text{rad/s}$?

(A) $-60°$ (B) $-35°$ (C) $-24°$ (D) $-45°$

5. For the transfer function, $H(s) = \dfrac{200\,s}{1+0.01\,s}$, what is the phase obtained using the Bode approximation at $\omega = 500\,\text{rad/s}$?

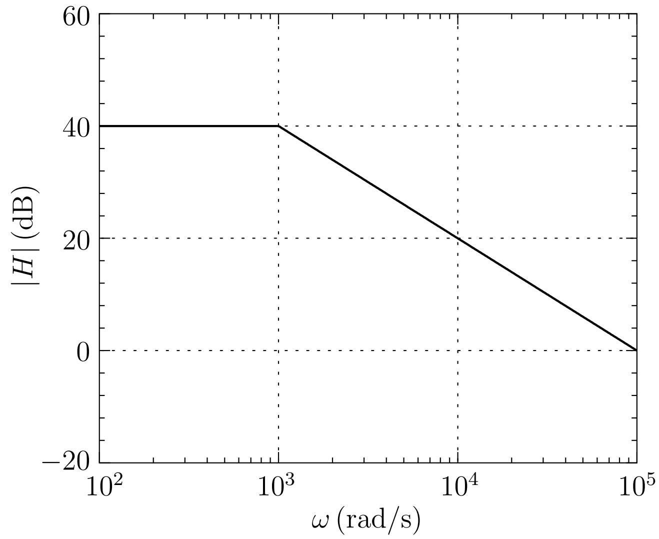
(A) 16.2° (B) 11.3° (C) 13.5° (D) 15.5°

6. The impedance $Z(s)$ in the figure is given by

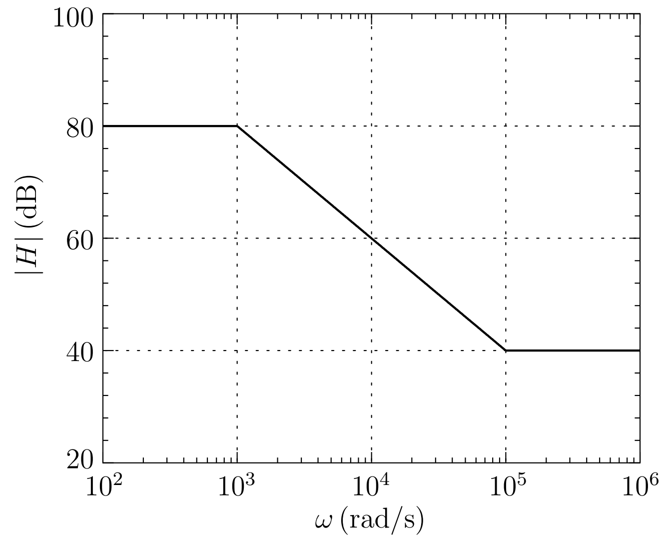


(A) $\dfrac{R_1}{1+sR_1C}$ (B) $R_2\dfrac{1+sR_1C}{1+sR_2C}$ (C) $\dfrac{R_1}{1+sR_2C}$ (D) $\dfrac{1+sR_2C}{1+sR_1C}$

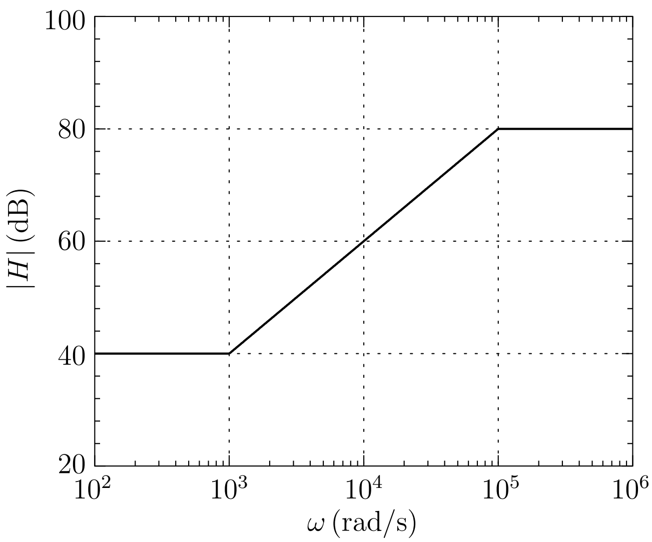
7. Which of the following graphs shows the correct Bode magnitude plot for the transfer function $H(s) = 10^2 \dfrac{1 + s/10^3}{1 + s/10^5}$?



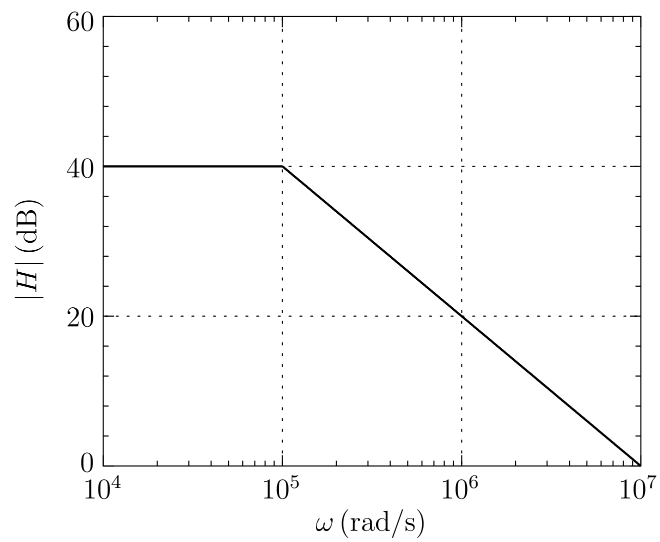
(A)



(B)

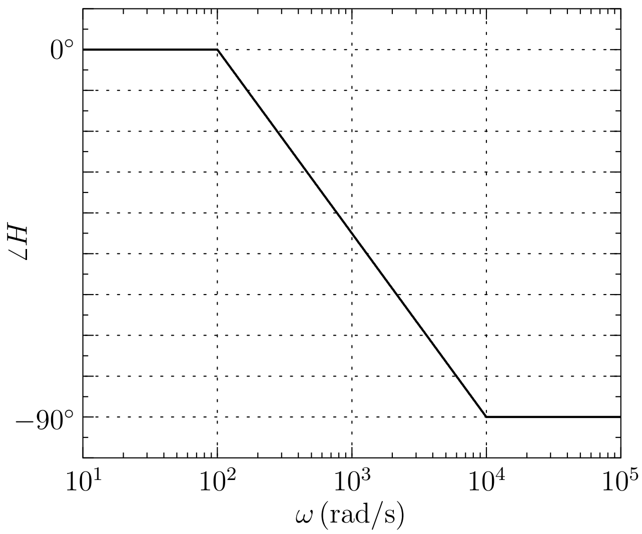


(C)

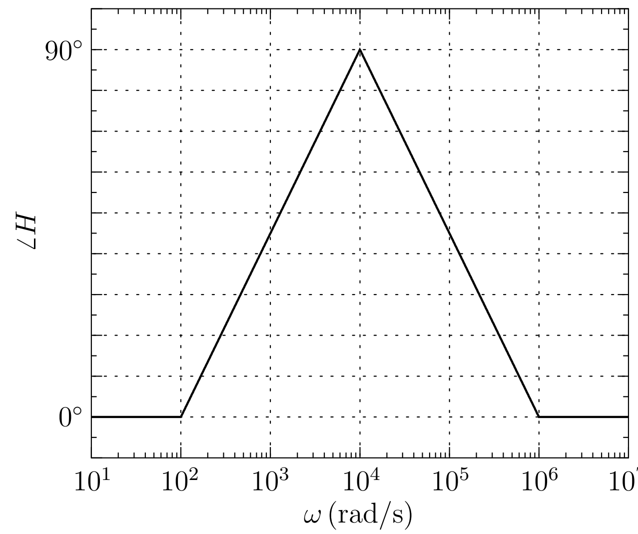


(D)

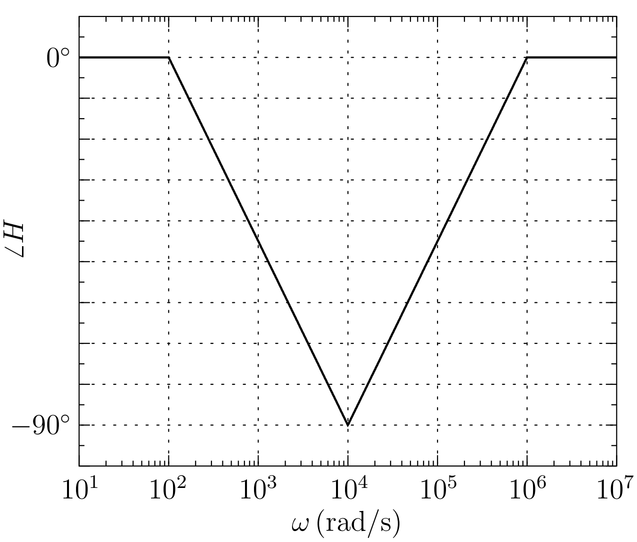
8. Which of the following graphs shows the correct Bode phase plot for the transfer function $H(s) = 10^2\,\dfrac{1 + s/10^3}{1 + s/10^5}$?



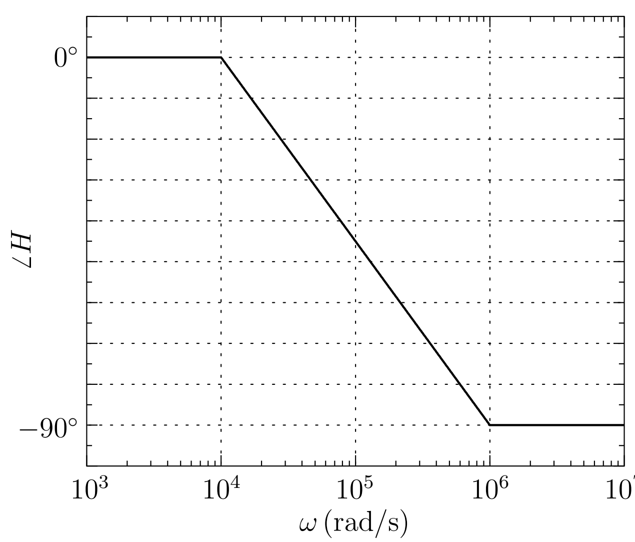
(A)



(B)

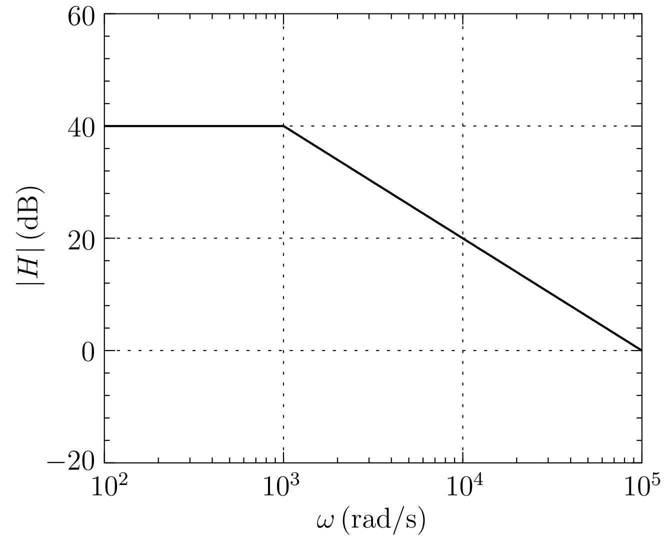


(C)

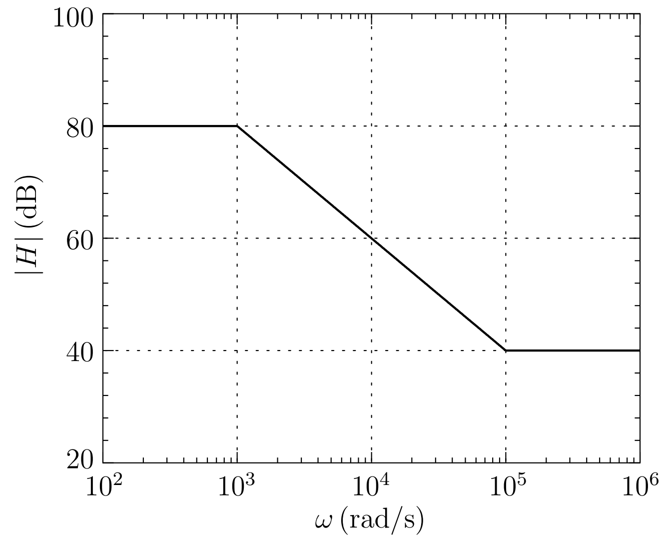


(D)

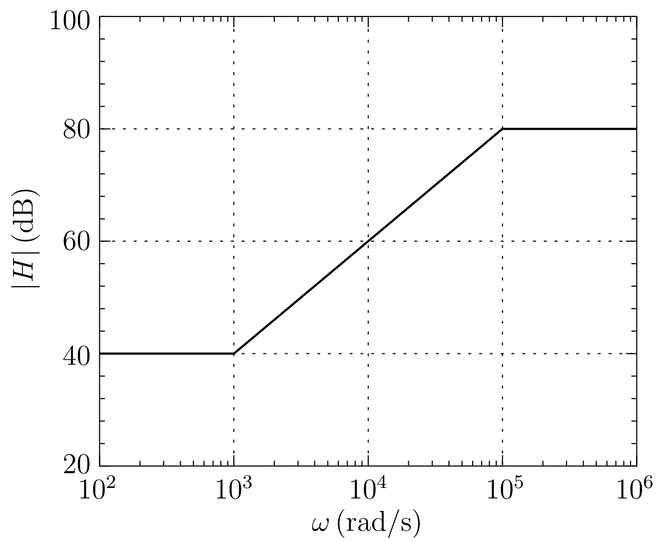
9. Which of the following graphs shows the correct Bode magnitude plot for the transfer function $H(s) = 10^4 \, \dfrac{1 + s/10^5}{1 + s/10^3}$?



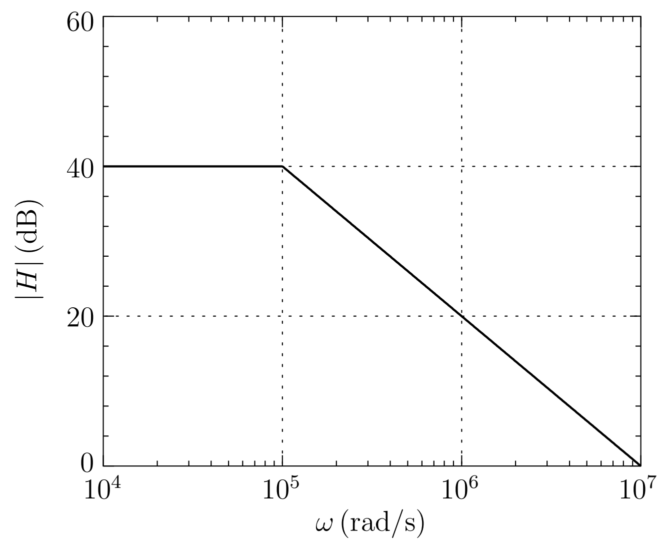
(A)



(B)

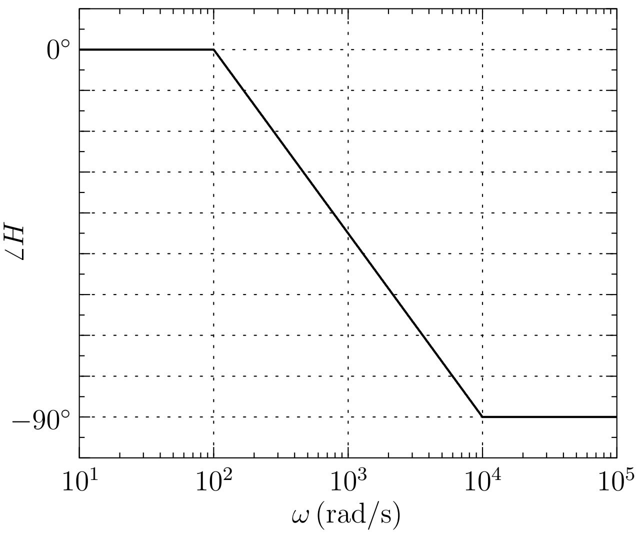


(C)

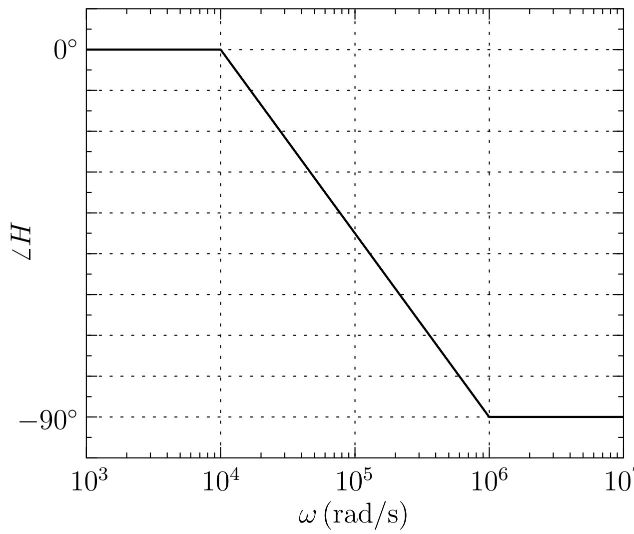


(D)

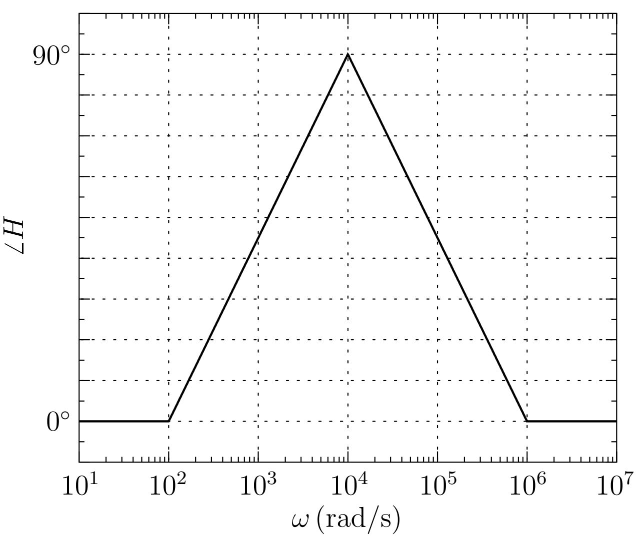
10. Which of the following graphs shows the correct Bode phase plot for the transfer function $H(s) = 10^4 \dfrac{1 + s/10^5}{1 + s/10^3}$?



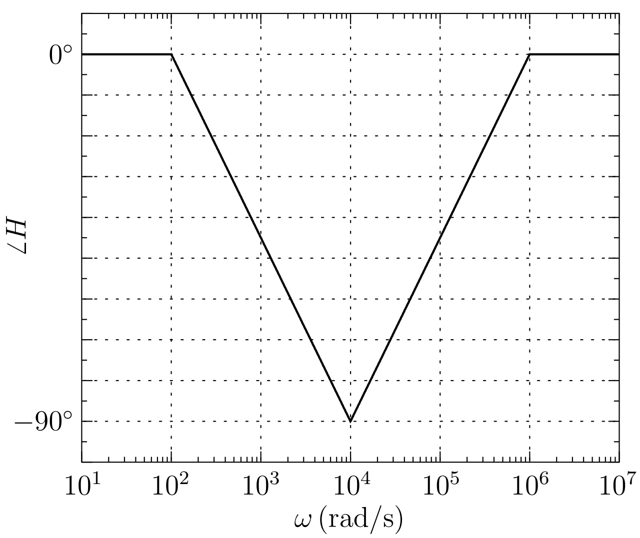
(A)



(B)

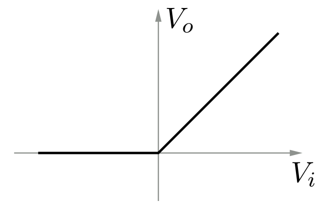


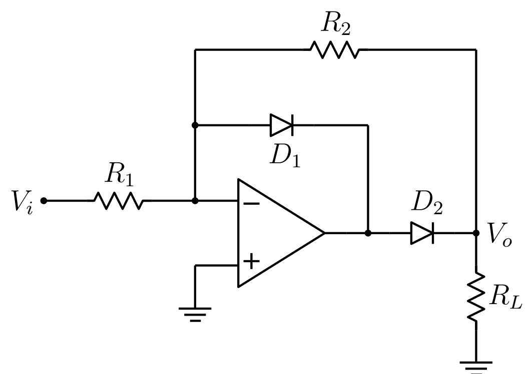
(C)



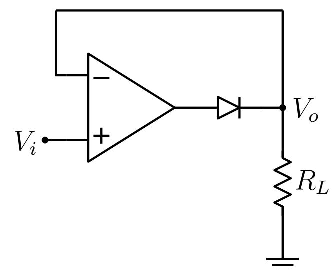
(D)

11. Which of the following circuits will produce the V_o versus V_i relationship shown in the figure?

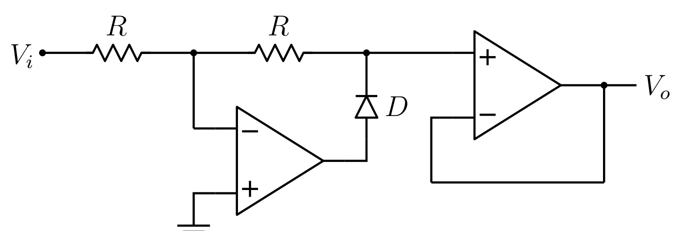




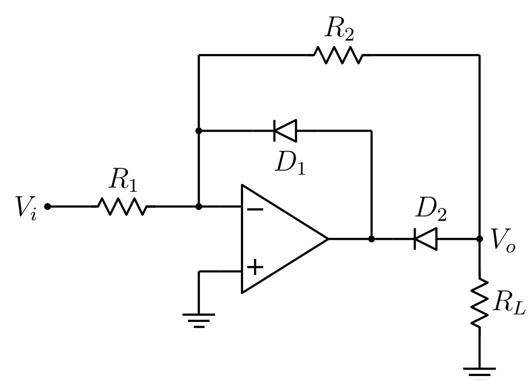
(A)



(B)

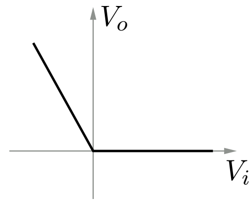


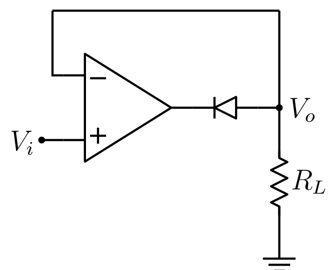
(C)



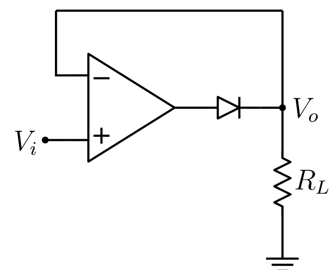
(D)

12. Which of the following circuits will produce the V_o versus V_i relationship shown in the figure?

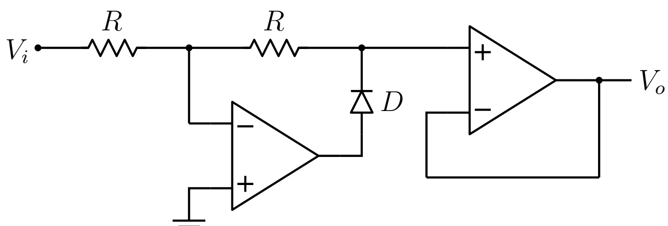




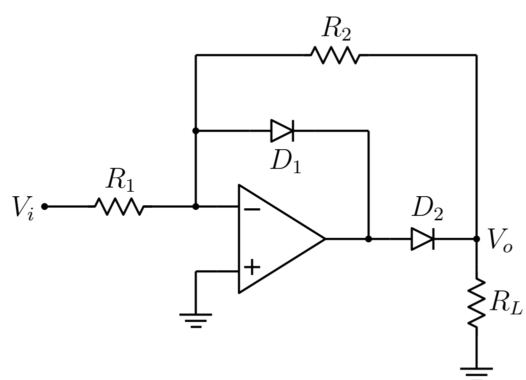
(A)



(B)

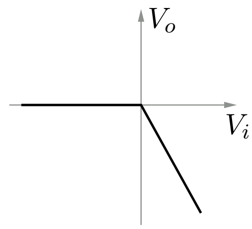


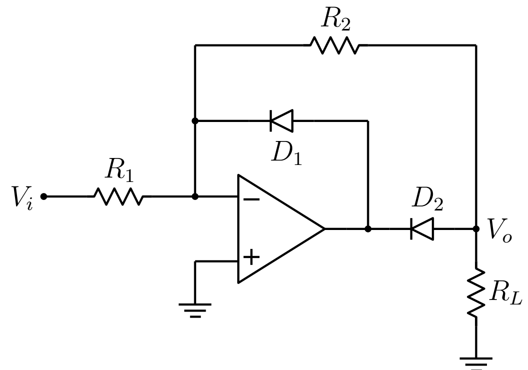
(C)



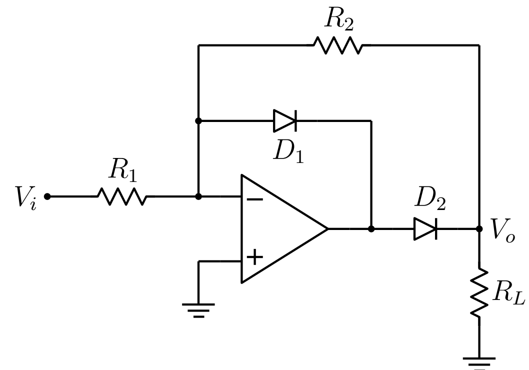
(D)

13. Which of the following circuits will produce the V_o versus V_i relationship shown in the figure?

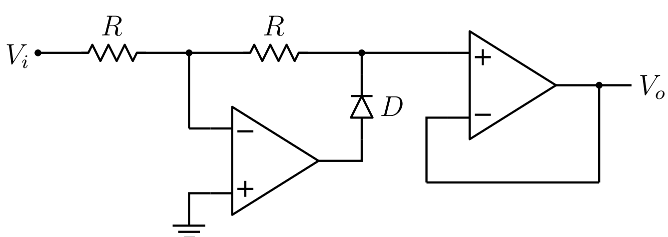




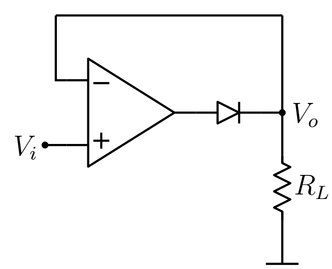
(A)



(B)

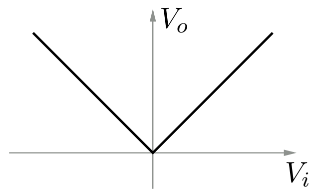


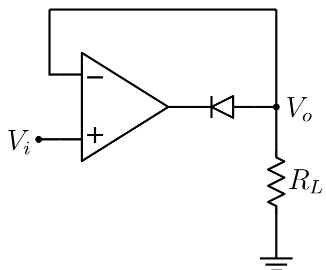
(C)



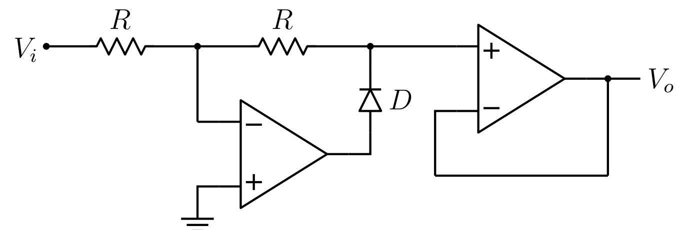
(D)

14. Which of the following circuits will produce the V_o versus V_i relationship shown in the figure?

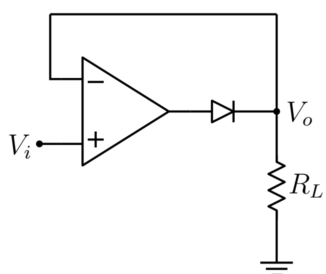




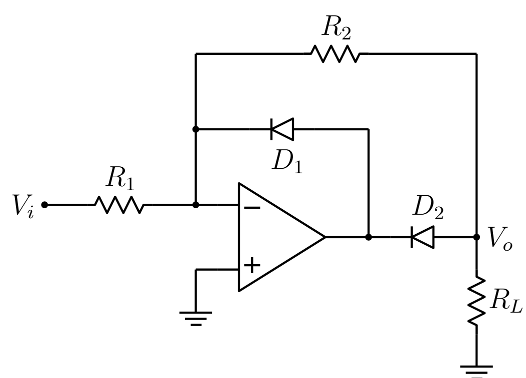
(A)



(B)

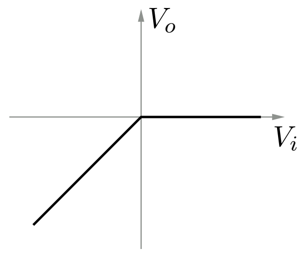


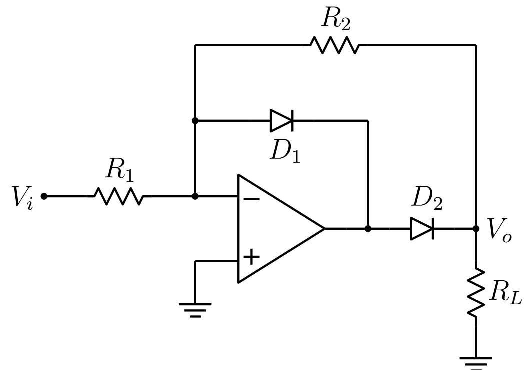
(C)



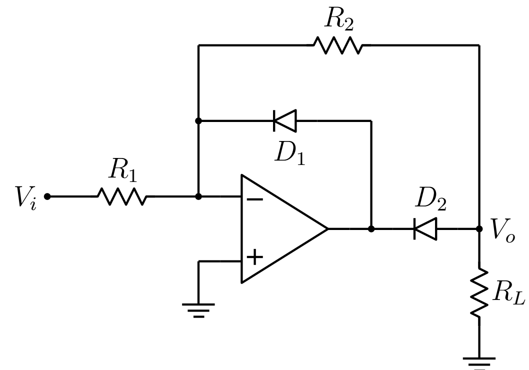
(D)

15. Which of the following circuits will produce the V_o versus V_i relationship shown in the figure?

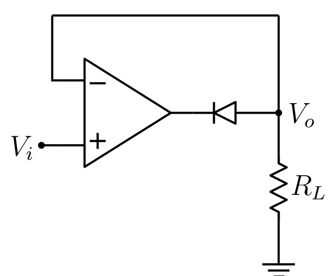




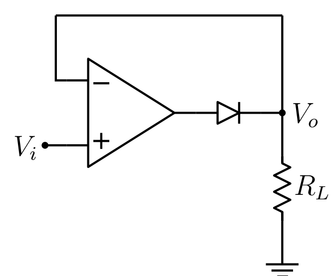
(A)



(B)



(C)



(D)